The ongoing MAMELI study: MApping the Methylation of repetitive elements to track the Exposome effects on health: the city of Legnano as a LIving lab

Nardi Tiago[1], Pandolfini Luca[2], Iodice Simona[1], Favero Chiara[1], Fanti Giacomo[1], Biganzoli Elia[3], Bollati Valentina[1]

1 Department of Clinical Sciences and Community Health, University of Milan
2 Center for Human Technologies, Istituto Italiano di Tecnologia (IIT), Genova
3 Department of Biomedical and Clinical Sciences, University of Milan

UNIVERSITÀ DEGLI STUDI DI MILANO

Background

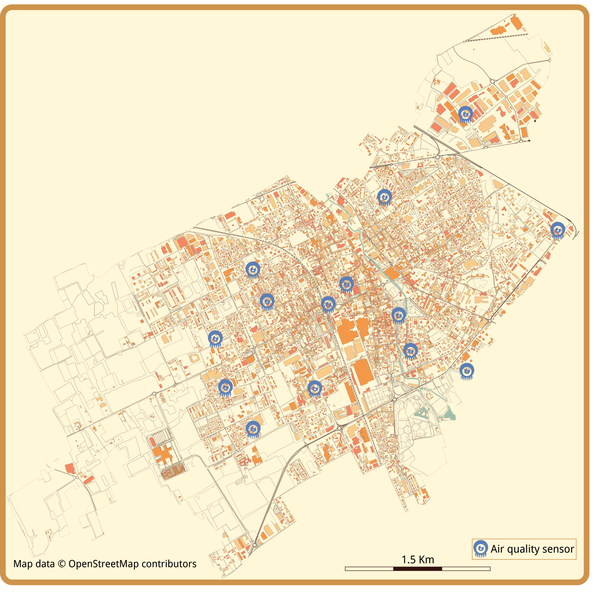
Several studies investigated the effects of harmful **exposures** (e.g. air pollutants) looking at epigenetic modifications.
In particular, changes in the **methylation** of **transposable elements** (TEs), such as LINE-1, have been linked to many environmental **exposures**, with the assumption that these alterations are inherently pathogenic

Our Hypothesis

"**Responsive TEs"** exapted as part of adaptive response to the **exposome,** the set of disease determinants during the life course:
limited epigenetic modification in responsive TEs would impair the adaptation to the exposome[1]

Who and where

City of Legnano: 60 000 residents
6200 to recruit for **MAMELI**:
- 200 from blood donors
- 6000 from the general population



How

Identifying "**Responsive TEs**", with the city of Legnano as a living lab, monitoring the **exposome** of a cohort of 6200 residents

3 Phases:
- Discovery: Identify **TEs with differential methylation** in 200 blood donors in 2 times (6 months)
- Tuning: Measuring and modelling the dependence between **TEs methylation** and **exposome** in 2500 subjects
- Assessing: Testing the **Exposome** / **TE-methylation** model in 3500 subjects

Workflow for Responsive TEs Identification

	Timeline
Discovery: Nanopore sequencing 200 subjects	Starting date: 2024-02
- Entire genome: DNA sequences + **5mC**	
- Selection of **TEs**	
- Selection of **TEs** with methylation changes between T1 and T2	Expected ending date: 2024-12
Tuning: methylation panel on 2500 subjects	Expected starting date: 2024-12
- **TEs** with methylation associated with **exposome** factors	Expected ending date: 2026-05
Assessing: methylation panel on 3500 subjects	Expected starting date: 2026-06
- Measuring **TEs** methylation	
- Identification of subjects with impaired TEs response	Expected ending date: 2027-12

1: Bollati V, Biganzoli EM, Carugno M. What if ... ? A new hypothesis to approach the relationship between environmental stimuli, biological features, and health. Heliyon. 2023 Nov 28;9(12):e22985. doi: 10.1016/j.heliyon.2023.e22985. PMID: 38058436

The Exposome:

Subjects **monitored** 2 weeks before blood/urine sampling:

- **Air Quality**
- **Physiological data**
- **Pollutants**
- **Lifesyle**

Air Quality

14 **sensors** ( )in the city of Legnano +
Regional Air Quality Data (ARPA) +
GPS locations of each subject

Physiological data

Wearable device measuring
Heart Rate (Variability),
Sleep Parameters,
Physical Activity,
Breathing rate

Pollutants

Mass Spectrometry to assay:
- Urinary Metals
- Polycyclic aromatic hydrocarbons
- Phthalate metabolites
- PFASs
- Occupational toxicants

Lifestyle

Questionaries and personal app to track diet, habits (smoking, alcohol consumption)

Contacts
Visit us at **mameli.unimi.it**
Contact us at **mameli@unimi.it**

Funded by the European Union
European Research Council
Established by the European Commission